

Convertible Bond Pricing: Lattice Tree and Least Squares Monte Carlo Methods

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December 15, 2025

Abstract

This paper investigates the application of numerical methods for pricing convertible bonds, a class of hybrid securities that exhibit both debt and equity characteristics. We implement and analyze two principal valuation frameworks: the Least-Squares Monte Carlo (LSMC) simulation and lattice-based tree models. The study begins by validating these methods on simpler European, American, and Asian options, establishing their accuracy and computational performance. A key contribution of this work is the application of importance sampling as a variance reduction technique to improve the efficiency of Monte Carlo simulations, particularly for options that are deep out-of-the-money. The analysis is then extended to price convertible bonds, incorporating features such as discrete coupon payments and dividend yields. The results from both the LSMC and lattice methods are compared to assess their relative strengths and weaknesses in handling the complex early-exercise features of convertible bonds. This research demonstrates the practical implementation of these advanced computational techniques in financial engineering and provides a quantitative basis for their use in complex derivative pricing.

1 Part 1

1.1 Introduction

The valuation of financial derivatives is a cornerstone of modern quantitative finance. While analytical solutions like the Black-Scholes model (AI Quant, 2023) provide closed-form prices for simple options, more complex instruments, particularly those with early-exercise features or path-dependencies, require robust numerical methods. This paper explores two such computational frameworks: Monte Carlo simulation (Park, 2023) and lattice tree models, with a specific focus on their application to pricing convertible bonds.

To analyze the price of convertible bonds and their optimal exercise boundaries, we first develop and validate these techniques on standard option types. Part 1 of this report details the implementation of Monte Carlo methods for European call, put, and Asian options, with a focus on convergence analysis and variance reduction through importance sampling. It also introduces binomial and trinomial lattice models. Part 2 applies these developed techniques to the more complex case of convertible bonds, demonstrating how they can be adapted to handle the instrument's hybrid nature. Through this structured approach, we aim to provide a comprehensive analysis of the efficacy of these methods in derivative pricing.

In financial mathematics, calculating the true price of derivatives, such as options, can be challenging due to the complexity of the underlying asset dynamics. One powerful method for approximating option prices is the *Monte Carlo simulation* (Park, 2023; Reid Research, 2023), a computational algorithm that relies on repeated random sampling to obtain numerical results. This paper explores the application of Monte Carlo simulation to approximate the prices of call options, put options, and Asian options, analyzing how the price and associated error vary with the number of simulations.

The price dynamics of the underlying asset is given by the stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where S_t is the asset price at time t , μ is the drift, and σ is the volatility. Under the risk-neutral measure, μ is replaced by the risk-free interest rate r . The solution to this SDE is

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right)$$

where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

To price European options under geometric Brownian motion (GBM) using Monte Carlo, the algorithm to estimate V by Monte Carlo simulation is:

1. Generate a set of sample normals $Z_p \sim \mathcal{N}(0, 1)$ for $p = 1, \dots, N$, using the inverse transform method.
2. For each Z_p , discount back:

$$Y_p = e^{-rT} f(S_T^p)$$

where

$$S_T^p = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z_p \right).$$

3. By the law of large numbers,

$$V \approx \frac{1}{N} \sum_{p=1}^N Y_p.$$

1.2 Parameter Setting

We assume $T = 1$ year throughout unless otherwise specified.

1.3 Call Option Pricing

We consider a European call option with the following parameters:

- Initial stock price: $S_0 = 60$

- Strike price: $K = 65$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 1 billion. The results, including the estimated option price, standard error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 1,000,000,000$ due to excessive time duration and storage requirements.

Table 1: Monte Carlo Simulation Results for Call Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration (s)
1,000	5.6934272154	0.3524039067	6.1896621025404476	0.0416683750
10,000	6.0885754219	0.1170608722	1.922631552392815	0.0293462910
100,000	5.8368203180	0.0365350242	0.6259405330035073	0.0378850839
200,000	5.8966270577	0.0258306067	0.4380573244412722	0.0136703750
1,000,000	5.8979124817	0.0116255343	0.1971126951820443	0.0800332080
10,000,000	5.9015376269	0.0036732493	0.06224224188946671	0.4007488750
100,000,000	5.8987551805	0.0011606359	0.00019675945799149642	4.8253777500
1,000,000,000	5.8992701404	0.0003670859	0.0000622256	331.601870

1.4 Analysis

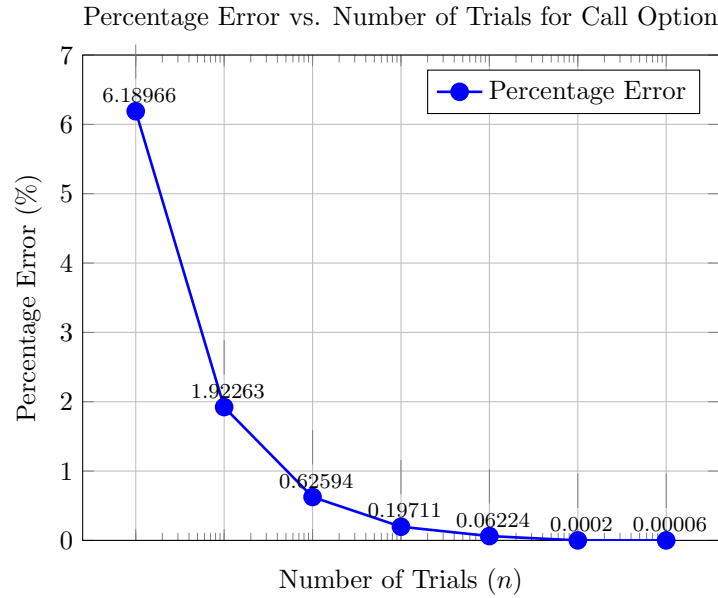


Figure 1: Relationship between the number of trials (n) and percentage error for call option pricing using Monte Carlo simulation, with labeled coordinates.

The simulation results indicate that at $n = 1,000,000$, the percentage error decreases to approximately 0.197%, which is well below the target threshold of 0.5%. This suggests that $n = 1,000,000$ provides a sufficiently accurate approximation for the call option price under the given parameters. Therefore, we fix $n = 1,000,000$ for further analysis to balance accuracy and computational efficiency.

Future work will extend the Monte Carlo simulation to put options and Asian options, comparing their price approximations and error behaviors under similar conditions.

1.5 Put Option Pricing

We consider a European put option with the following parameters:

- Initial stock price: $S_0 = 60$
- Strike price: $K = 55$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 1 billion. The results, including the estimated option price, standard error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 1,000,000,000$ due to excessive time duration and storage requirements.

Table 2: Monte Carlo Simulation Results for Put Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration (s)
1,000	3.4977868700	0.1874162840	5.3581390440023594	0.0112356250
10,000	3.9665790303	0.0646841349	1.6307285038334316	0.0587356660
100,000	3.9429362680	0.0203820351	0.5169252994599188	0.0357832500
1,000,000	3.9458729605	0.0064305971	0.16297020139723639	0.0890570000
10,000,000	3.9467271516	0.0020326618	0.05150246566396758	0.4012099589
100,000,000	3.9505680591	0.0006432265	0.016284699418928474	5.1449937921
1,000,000,000	3.9498083425	0.0002033819	0.0000514916	350.4375698000

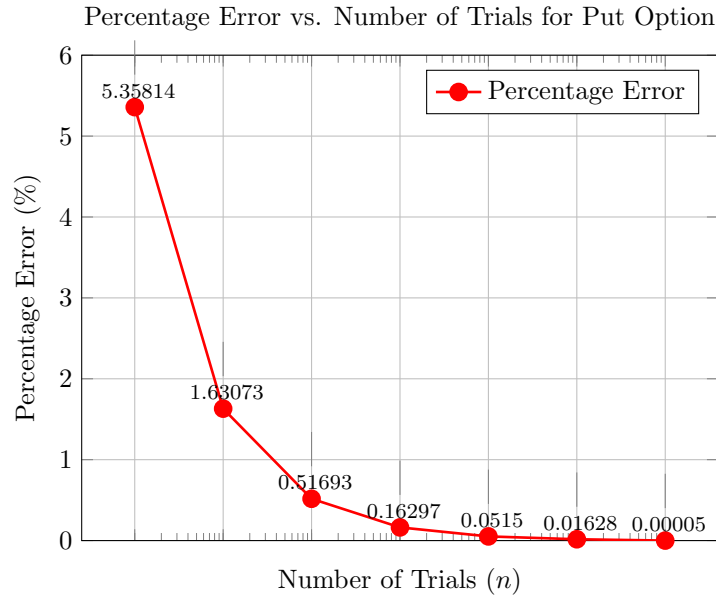


Figure 2: Relationship between the number of trials (n) and percentage error for put option pricing using Monte Carlo simulation, with labeled coordinates.

1.6 Asian Option Pricing

We consider an Asian call option with the following parameters:

- Initial stock price: $S_0 = 60$
- Strike price: $K = 65$
- Risk-free rate: $r = 3\%$ (0.03)
- Volatility: $\sigma = 0.3$
- Number of steps: $m = 365$

The Monte Carlo simulation is performed for varying numbers of trials (n), ranging from 1,000 to 10 million. The results, including the estimated option price, standard error, percentage error, and time duration, are summarized in the table below. Note that simulations were stopped at $n = 10,000,000$ due to excessive time duration and storage requirements.

Table 3: Monte Carlo Simulation Results for Asian Option Pricing

Number of Trials (n)	Price	Standard Error	Percentage Error (%)	Time Duration (s)
1,000	2.9493238862	0.1873925776	6.353746988893269	0.3233828751
10,000	2.4778316263	0.0546414057	2.2052106004030313	3.1388878340
100,000	2.5406059852	0.0174188498	0.6856179166712045	31.6894467090
1,000,000	2.5618290434	0.0055063970	0.21494006516673218	309.2048262920
10,000,000	2.5573364649	0.0017419616	0.06811624569485263	3344.7093135830

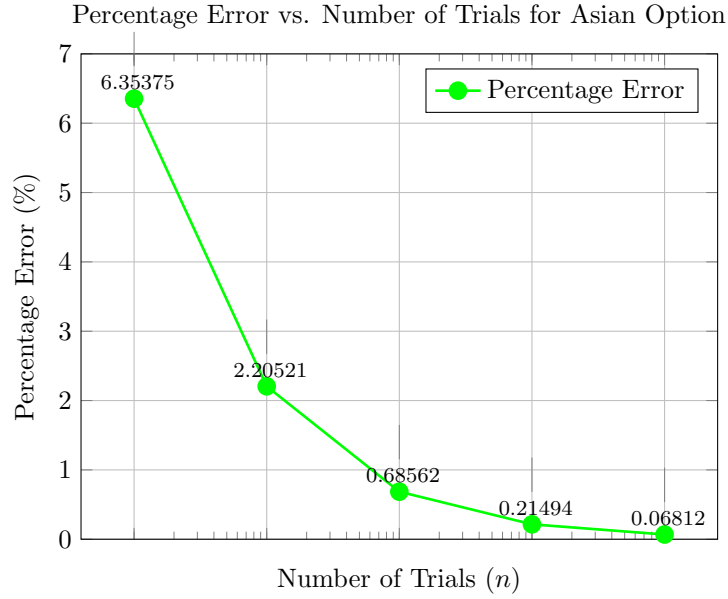


Figure 3: Relationship between the number of trials (n) and percentage error for Asian option pricing using Monte Carlo simulation, with labeled coordinates.

1.7 Relationship Between the Strike Price and the Percentage Error in Monte Carlo Simulation

To further investigate the behavior of the Monte Carlo simulation, we fix the number of trials at $n = 200,000$ and analyze how the percentage error of the call option price varies with changes in the strike price (K). By varying the strike price while keeping the initial stock price ($S_0 = 60$), risk-free rate ($r = 0.03$), and volatility ($\sigma = 0.3$) constant, we aim to understand the sensitivity of the pricing error to different strike price levels. This analysis provides insights into the robustness of the Monte Carlo simulation across a range of option moneyness conditions.

Table 4: Relation Between Strike Price K and the Error for $n = 200,000$ for Call Option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.9030471319	0.0260105784	0.44062969182064775
70	4.3108548296	0.0226688466	0.5258550218202654
80	2.2085308848	0.0166125454	0.7521989177289221
100	0.5045578207	0.0079014707	1.5660188782655925
120	0.1235174742	0.0041755108	3.380502102142773
145	0.0155736815	0.0013123009	8.426401459816489

Percentage Error vs. Strike Price for $n = 200,000$ (Call Option)

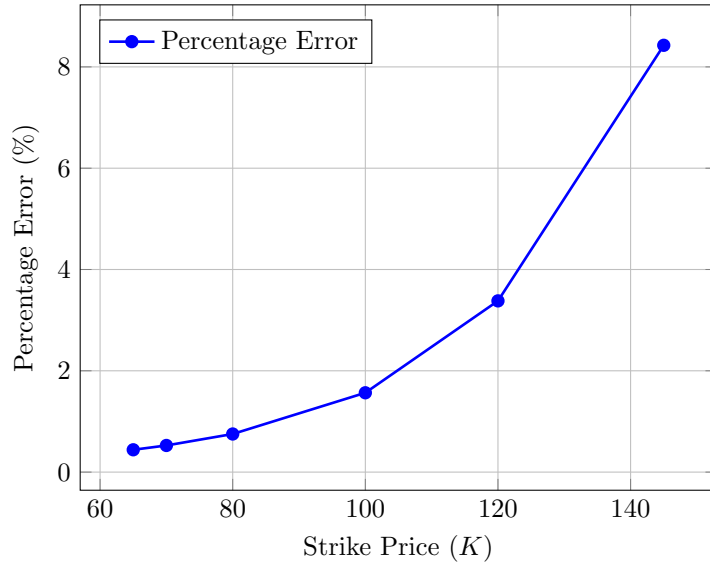


Figure 4: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$ (Call Option).

Table 5: Relation Between Strike Price K and the Error for $n = 200,000$ for Put Option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9380010143	0.0143560978	0.3645529239890197
50	2.2485225882	0.0104507612	0.464783466881606
40	0.4652423990	0.0041779326	0.8980120035938425
30	0.0347791649	0.0009082143	2.6113747848661533
25	0.0037966588	0.0002453879	6.463259689570557
20	0.0001852903	0.0000579096	31.253457061023754

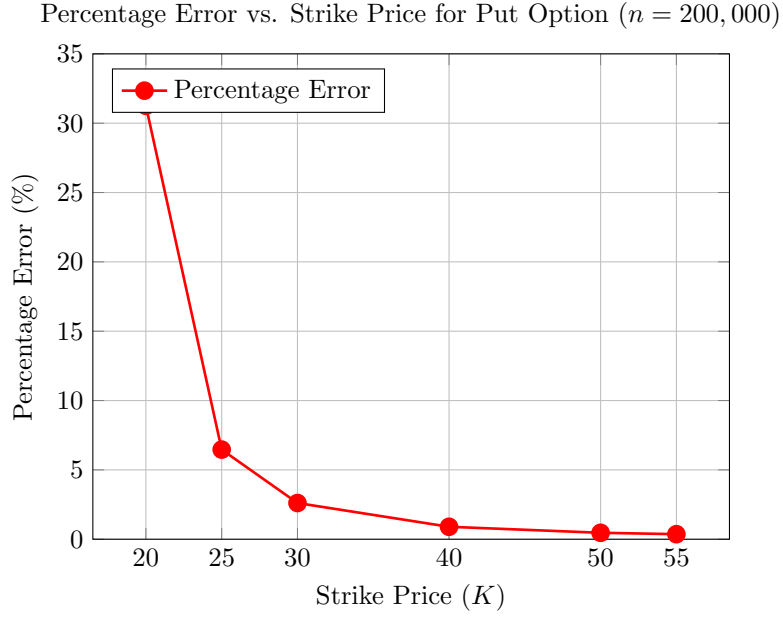


Figure 5: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$ (Put Option).

Table 6: Relation Between Strike Price K and the Error for $n = 200,000$ for Asian Option

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	2.5574461955	0.0122824000	0.4802603464888412
70	1.3640460181	0.0091348157	0.6696853009332701
80	0.3259984252	0.0044256591	1.3575707059465612
100	0.0137401286	0.0009628217	7.0073703249731
120	0.0006620147	0.0002653571	40.08326233750248



Figure 6: Relationship between strike price (K) and percentage error for Asian option pricing with $n = 200,000$, with labeled coordinates.

1.8 Reasons for Poor Accuracy

When the strike price of an option increases significantly (for calls) or decreases significantly (for puts), the percentage error in option pricing becomes unacceptably high. This raises the question: what causes this issue, and how can we improve the accuracy? To address this, we employ a well-known technique called importance sampling (Masoud, 2023).

Before introducing importance sampling, we investigate why the accuracy deteriorates for deep out-of-the-money options. Consider the call option price using the risk-neutral expectation:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)],$$

which can be expressed as:

$$V_0 = e^{-rT} [\mathbb{E}[\max(S_T - K, 0) \mid S_T > K]P(S_T > K) + \mathbb{E}[\max(S_T - K, 0) \mid S_T < K]P(S_T < K)].$$

Since $\max(S_T - K, 0) = 0$ when $S_T < K$, the second term vanishes, simplifying to:

$$V_0 = e^{-rT} P(S_T > K) \mathbb{E}[\max(S_T - K, 0) \mid S_T > K].$$

The probability $P(S_T > K)$ is the expectation of the indicator $1_{\{S_T > K\}}$. However, the conditional expectation $\mathbb{E}[\max(S_T - K, 0) \mid S_T > K]$ requires generating stock prices $S_T > K$. Since S_T follows a lognormal distribution (via Brownian motion), generating rare large S_T values becomes difficult as K increases, leading to high variance and poor accuracy in Monte Carlo estimates. Importance sampling addresses this by shifting the sampling distribution to favor rare events.

1.9 Normal Distribution of Stock Price

To illustrate the challenge of generating $S_T > K$ in Monte Carlo simulations, consider the (log) distribution of S_T , which is normal. The figure below shows the standard normal density with mean at 0 (corresponding to the log-stock price mean) and K to the right, highlighting the tail probability $P(S_T > K)$. The shifted distribution (mean at K) increases the likelihood in the tail.

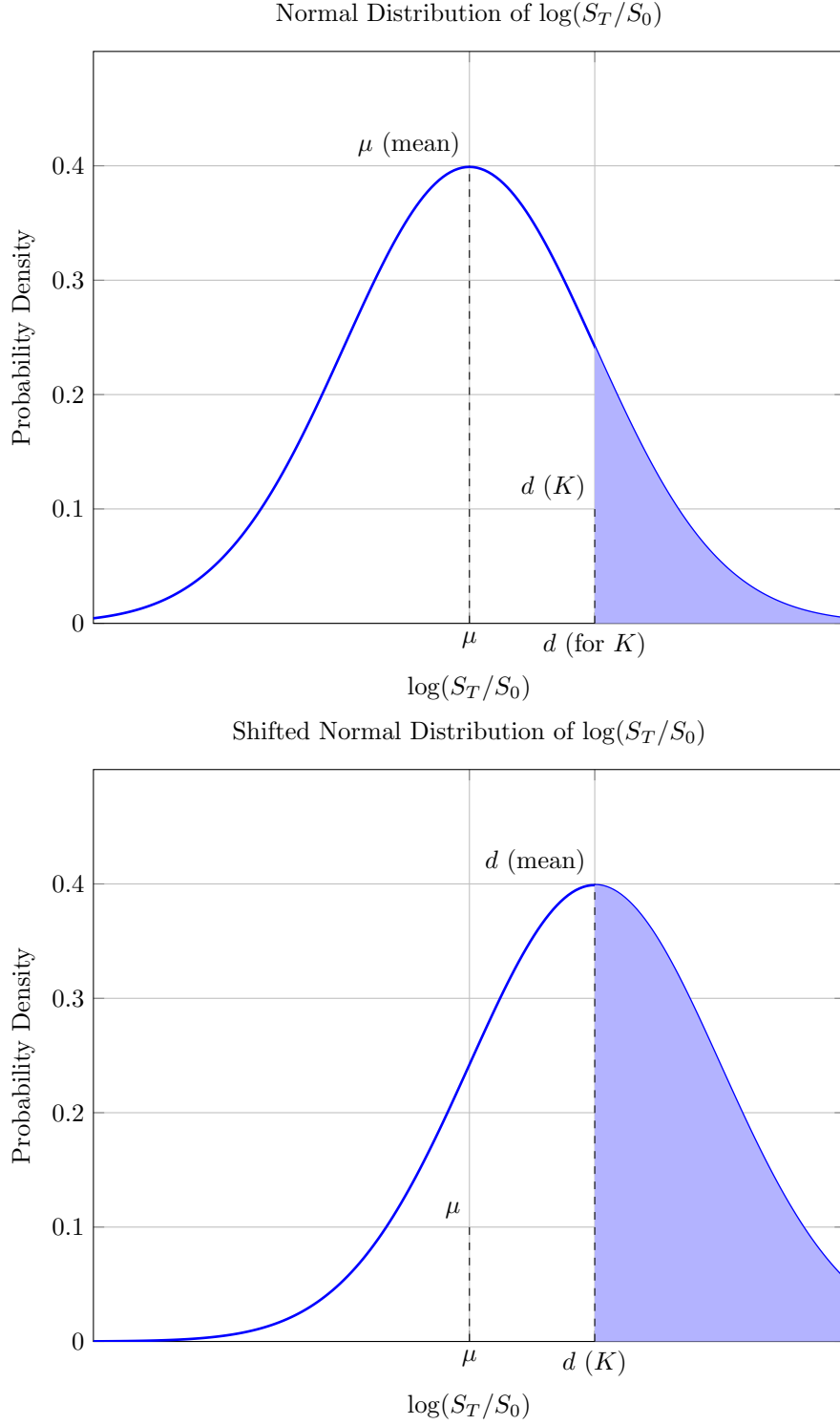


Figure 7: Top: Normal distribution of $\log(S_T/S_0)$ with mean $\mu = (r - \sigma^2/2)T$. Bottom: Shifted normal distribution with mean at $d \approx \log(K/S_0)$. Both show the shaded tail area $P(\log(S_T/S_0) > d)$, which has higher probability in the shifted case for generating $S_T > K$.

1.10 Importance Sampling (Change of Measure)

The core idea of importance sampling (Masoud, 2023) is to generate more samples where $S_T > K$ (for calls) by changing the sampling measure. Originally, $\log(S_T/S_0) \sim \mathcal{N}(\mu, \sigma^2 T)$ with $\mu = (r - \sigma^2/2)T$. The expectation is:

$$\mathbb{E}_f[h(S_T)] = \int h(x)f(x) dx,$$

where f is the original density. We introduce a new density $g(x)$ (shifted normal $\mathcal{N}(\mu + \theta, \sigma^2 T)$ for some shift $\theta > 0$) and rewrite:

$$\mathbb{E}_f[h(S_T)] = \mathbb{E}_g \left[h(x) \frac{f(x)}{g(x)} \right].$$

The likelihood ratio $f(x)/g(x) = \exp(-\theta(\log x - \mu)/(\sigma^2 T) + \theta^2/(2\sigma^2 T))$ corrects for the change. We choose $\theta \approx \log(K/S_0) - \mu$ to center the mass in the tail.

The classical pricing formula is $S_T = S_0 \exp((r - \sigma^2/2)T + \sigma\sqrt{T}Z)$, with $Z \sim \mathcal{N}(0, 1)$. Under g , $Z \sim \mathcal{N}(\theta/(\sigma\sqrt{T}), 1)$.

Originally, with $n = 200,000$, errors were high for extreme K . With importance sampling, errors are drastically reduced, as shown below.

Table 7: Relation Between Strike Price K and the Error for $n = 200,000$ for Call Option After Importance Sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.8818571310	0.0070251153	0.1194370280128
70	4.3016414843	0.0056079801	0.1303683739831
80	2.1986407229	0.0034892939	0.1587023231909
100	0.5246847674	0.001246	0.2380389366945
120	0.1178213526	0.0004209657	0.3572915221543
145	0.0179250022	0.0001055605	0.5889010725830

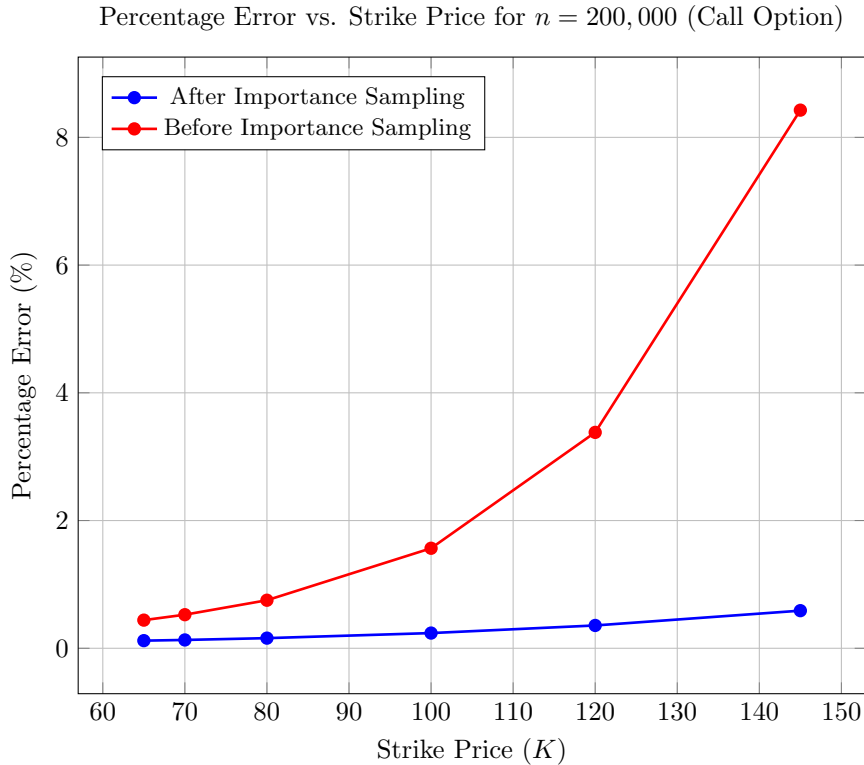


Figure 8: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$ (Call Option).

Table 8: Relation Between Strike Price K and the Error for $n = 200,000$ for Put Option After Importance Sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9504380596	0.0043639532	0.1104675775744
50	2.2664069060	0.0028716932	0.1267068667896
40	0.4706504711	0.0008745451	0.1858162575359
30	0.0325958339	0.0001135192	0.3482628384181
25	0.0040788935	0.0000228700	0.5606901009596
20	0.0002017843	0.0000022716	1.1257547338531

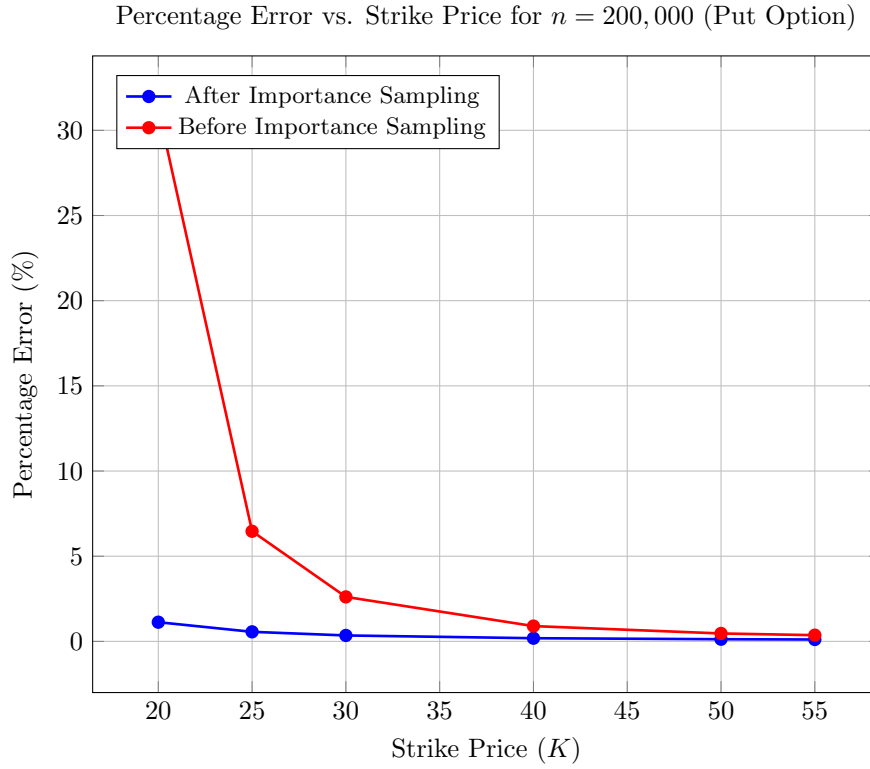


Figure 9: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$ (Put Option).

1.11 Path-Dependent Options with Importance Sampling

Now, we apply importance sampling (Masoud, 2023) to path-dependent Asian options (Helali, 2023) with $n = 1,000,000$. For Asian options, the shift is applied to the Brownian increments to favor paths where the average exceeds K .

Table 9: Relation Between Strike Price K and the Error for $n = 1,000,000$ for Call Option After Importance Sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.9014681454	0.0031470324	0.0533262623523
70	4.2964380803	0.0025093720	0.0584058699562
80	2.1992973271	0.0015607173	0.0709643624772
100	0.5246762827	0.0005583837	0.1064244289256
120	0.1181249456	0.0001883408	0.1594419875788
145	0.0181125664	0.0000474354	0.2618922154102

Table 10: Relation Between Strike Price K and the Error for $n = 1,000,000$ for Put Option After Importance Sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9512231202	0.0019514837	0.0493893584183
50	2.2672691020	0.0012846926	0.0566625555659
40	0.4700305078	0.0003907590	0.0831348149654
30	0.0328826044	0.0000510000	0.1550972568203
25	0.0040565666	0.0000102098	0.2516859715651
20	0.0002018025	0.0000010149	0.5028973426857

Table 11: Relation Between Strike Price K and the Error for $n = 1,000,000$ for Asian Call Option After Importance Sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	2.5645609853251665	0.0037335023013196416	0.14558056223592836
70	1.370926588692837	0.002004978094734432	0.14624985110589734
80	0.333514790605139	0.0006583680852701578	0.1974029649706376
100	0.013667260376728172	0.00006185267663825915	0.4525609005267679
120	0.00045757335722987623	0.000005326766043550572	1.164133785192066
145	0.000006775516222498219	0.0000004524948221504276	6.678381503211678

2 Introduction to Lattice Tree Methods

Having explored Monte Carlo simulation for pricing path-independent and path-dependent options, we now introduce another prominent method: lattice tree methods, based on the binomial model. These methods approximate derivative prices using a discrete-time lattice of possible stock price paths, employing a backward induction approach under the risk-neutral measure. In the following sections, we describe the binomial and trinomial trees.

2.1 Binomial Tree

Based on the Black-Scholes model, we know that for the future value $V_T(S_T)$, the initial value is given by

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[V_T(S_T)],$$

where $\mathbb{Q}$ stands for the risk-neutral probability measure. The stock price S_T satisfies the stochastic differential equation

$$\frac{dS_t}{S_t} = (r - q) dt + \sigma dW_t^{\mathbb{Q}},$$

where $W_T \sim N(0, T)$. In other words,

$$S_T = S_0 e^{(r - q - \frac{\sigma^2}{2})T + \sigma W_T^{\mathbb{Q}}}.$$

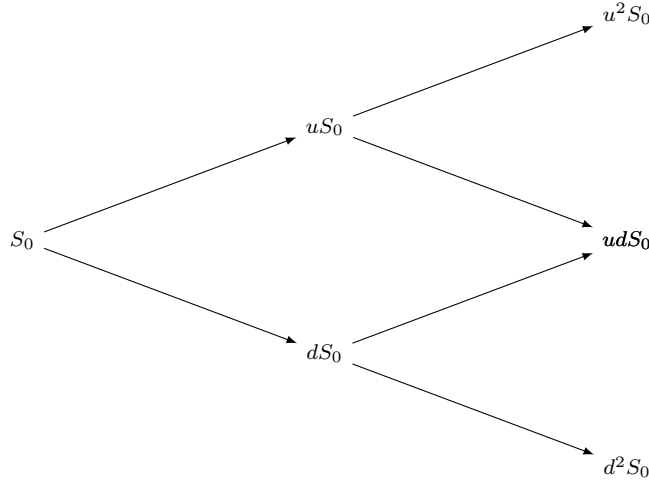


Figure 10: Binomial Tree for Two Periods

Hence, take the growth factors to be $u = e^v$ and $d = e^{-v}$, then denote $X_i = \text{either } v \text{ or } -v$ such that for n periods we have

$$S_T = S_0 e^{X_1 + X_2 + \dots + X_n} = S_0 e^{\sum_{i=1}^n X_i}.$$

We know that the Brownian motion follows a normal distribution; therefore, we have the mean $\mu = (r - q - \sigma^2/2)T$ and variance $\sigma^2 T$.

Revisiting the formula derived earlier, we have

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2 + \dots + X_n] = n\mathbb{E}[X_i],$$

where the X_i are independent and identically distributed.

We have

$$(r - q - \sigma^2/2)T = n\mathbb{E}[X_i] = nv(q_u - q_d),$$

where

$$q_u = \frac{e^{r\Delta t} - d}{u - d}, \quad u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}.$$

Hence, take the growth factors to be $u = e^v$ and $d = e^{-v}$. Then, denote X_i as either v or $-v$ such that for n periods, we have

$$S_T = S_0 e^{X_1 + X_2 + \dots + X_n} = S_0 e^{\sum_{i=1}^n X_i}.$$

We know that the Brownian motion follows a normal distribution; therefore, we have the mean $\left(r - q - \frac{\sigma^2}{2}\right)T$ and variance $\sigma^2 T$.

Revisiting the formula derived earlier, we have

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2 + \dots + X_n] = n\mathbb{E}[X_i]$$

(since the X_i are independent and identically distributed).

We have

$$\left(r - q - \frac{\sigma^2}{2}\right)T = n\mathbb{E}[X_i] = nv(q_u - q_d),$$

3 Trinomial Tree

Now, let us move on to the trinomial tree (Zhou Wang, 2021; Doga, 2023). In reality, stock prices do not always only go up or down; sometimes they remain unchanged. Therefore, in the approximation of derivatives, we allow the stock price to go up, remain unchanged, or go down. To calculate the probabilities of these movements, we introduce a set of constraints.

Define the growth factor as

$$v = \lambda\sigma\sqrt{\Delta t},$$

where q_u , q_m , and q_d represent the probabilities of the stock price going up, remaining unchanged, and going down, respectively, satisfying

$$q_u + q_m + q_d = 1.$$

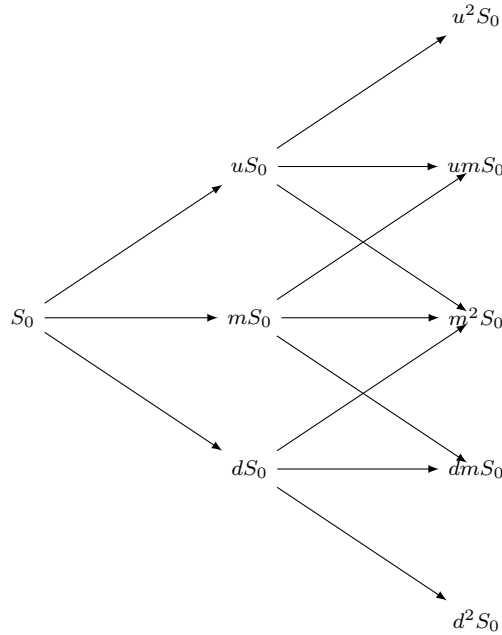


Figure 11: Trinomial Tree for Two Periods

Hence, we solve the system using algebraic methods to determine these probabilities.

In a trinomial tree model, we define the up, middle, and down factors for the stock price S at time step Δt as follows:

$$u = e^{\lambda\sigma\sqrt{\Delta t}}, \quad d = e^{-\lambda\sigma\sqrt{\Delta t}}, \quad m = 1, \quad (1)$$

$$S_u = Se^{\lambda\sigma\sqrt{\Delta t}}, \quad S_m = S, \quad S_d = Se^{-\lambda\sigma\sqrt{\Delta t}}, \quad (2)$$

where λ is a scaling parameter, σ is the volatility, and Δt is the time step.

The risk-neutral condition ensures the expected stock price matches the risk-free growth:

$$q_u u + q_m m + q_d d = e^{r\Delta t}, \quad (3)$$

where q_u , q_m , and q_d are the probabilities of the up, middle, and down moves, respectively, and r is the risk-free rate.

To match the variance of the stock price, we impose:

$$q_u(u - e^{r\Delta t})^2 + q_m(m - e^{r\Delta t})^2 + q_d(d - e^{r\Delta t})^2 = \sigma^2 \Delta t. \quad (4)$$

Substituting u , m , and d , and letting $v = \lambda\sigma\sqrt{\Delta t}$, we rewrite the equations:

$$q_u e^v + q_m \cdot 1 + q_d e^{-v} = e^{r\Delta t}, \quad (5)$$

$$q_u(e^v - e^{r\Delta t})^2 + q_m(1 - e^{r\Delta t})^2 + q_d(e^{-v} - e^{r\Delta t})^2 = \sigma^2 \Delta t. \quad (6)$$

Additionally, the probabilities must sum to one:

$$q_u + q_m + q_d = 1. \quad (7)$$

Using the substitution $v = \lambda\sigma\sqrt{\Delta t}$, we derive:

$$v(q_u - q_d) = \left(r - \frac{\sigma^2}{2}\right) \Delta t, \quad (8)$$

$$v^2(q_u + q_d) = \sigma^2 \Delta t. \quad (9)$$

These equations, along with $q_u + q_m + q_d = 1$, form the system:

$$\begin{pmatrix} v & 0 & -v \\ v^2 & 0 & v^2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \left(r - \frac{\sigma^2}{2}\right) \Delta t \\ \sigma^2 \Delta t \\ 1 \end{pmatrix}. \quad (10)$$

The determinant of the coefficient matrix is:

$$\det \begin{pmatrix} v & 0 & -v \\ v^2 & 0 & v^2 \\ 1 & 1 & 1 \end{pmatrix} = -2v^3. \quad (11)$$

The inverse of the coefficient matrix is:

$$\frac{1}{-2v^3} \begin{pmatrix} -v^2 & v & -v^2 \\ 0 & -2v^3 & 0 \\ -v^2 & -v & v^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2v} & \frac{1}{2v^2} & 0 \\ 0 & -\frac{1}{v^2} & 1 \\ -\frac{1}{2v} & \frac{1}{2v^2} & 0 \end{pmatrix}. \quad (12)$$

Solving for the probabilities:

$$\begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \frac{1}{2v} & \frac{1}{2v^2} & 0 \\ 0 & -\frac{1}{v^2} & 1 \\ -\frac{1}{2v} & \frac{1}{2v^2} & 0 \end{pmatrix} \begin{pmatrix} \left(r - \frac{\sigma^2}{2}\right) \Delta t \\ \sigma^2 \Delta t \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\left(r - \frac{\sigma^2}{2}\right) \Delta t}{2\lambda\sigma\sqrt{\Delta t}} + \frac{\sigma^2 \Delta t}{2(\lambda\sigma\sqrt{\Delta t})^2} \\ 1 - \frac{\sigma^2 \Delta t}{(\lambda\sigma\sqrt{\Delta t})^2} \\ -\frac{\left(r - \frac{\sigma^2}{2}\right) \Delta t}{2\lambda\sigma\sqrt{\Delta t}} + \frac{\sigma^2 \Delta t}{2(\lambda\sigma\sqrt{\Delta t})^2} \end{pmatrix}. \quad (13)$$

After simplification, the probabilities are:

$$\begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \frac{1}{2\lambda^2} + \frac{\left(r - \frac{\sigma^2}{2}\right)\sqrt{\Delta t}}{2\lambda\sigma} \\ 1 - \frac{1}{\lambda^2} \\ \frac{1}{2\lambda^2} - \frac{\left(r - \frac{\sigma^2}{2}\right)\sqrt{\Delta t}}{2\lambda\sigma} \end{pmatrix}. \quad (14)$$

4 Introduction to American Option Pricing Using a Trinomial Tree

Convertible bonds combine bond and equity features by allowing the holder to convert the bond into shares. The valuation requires modeling the embedded option and optimal conversion strategy. The Least Squares Monte Carlo (LSM) method (Longstaff Schwartz, 2001; Grant, 2020) simulates many stock price paths and uses regression to estimate the continuation value, enabling optimal exercise decision approximation.

Convertible bonds typically pay coupons continuously or at discrete intervals. In this model, we approximate continuous coupons by accruing small coupon payments at each simulation time step. This approach integrates coupon payments naturally into the backward induction algorithm, reflecting the time value of these cash flows throughout the life of the bond.

This document presents the LSM algorithm for fixed and time-dependent conversion ratios, including matrix visualizations to illustrate the process.

5 LSM Algorithm for Convertible Bonds with Fixed Conversion Ratio

5.1 Setup and Notation

- N : number of simulated paths
- M : number of time steps (with times $t = 0, \dots, M$)
- $S_{i,t}$: stock price for path i at time t
- $V_{i,t}$: bond value for path i at time t
- $D_{i,t}$: conversion decision (True if convert, False if hold)
- Fixed conversion ratio cr

5.2 Simulation Paths Matrix

The simulated stock prices for all paths and times can be organized as an $N \times (M + 1)$ matrix:

$$S = \begin{bmatrix} S_{1,0} & S_{1,1} & \cdots & S_{1,M} \\ S_{2,0} & S_{2,1} & \cdots & S_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ S_{N,0} & S_{N,1} & \cdots & S_{N,M} \end{bmatrix}$$

5.3 Bond Values Matrix

We similarly maintain an $N \times (M + 1)$ matrix of bond values, updated backward in time:

$$V = \begin{bmatrix} V_{1,0} & V_{1,1} & \cdots & V_{1,M} \\ V_{2,0} & V_{2,1} & \cdots & V_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ V_{N,0} & V_{N,1} & \cdots & V_{N,M} \end{bmatrix}$$

At maturity $t = M$, the bond value is:

$$V_{i,M} = \max(F, cr \times S_{i,M})$$

5.4 Conversion Decision Matrix

At each time step, we record the optimal conversion decision in an $N \times (M + 1)$ boolean matrix:

$$D = \begin{bmatrix} d_{1,0} & d_{1,1} & \cdots & d_{1,M} \\ d_{2,0} & d_{2,1} & \cdots & d_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N,0} & d_{N,1} & \cdots & d_{N,M} \end{bmatrix}$$

where

$$d_{i,t} = \begin{cases} \text{True,} & \text{if } cr \times S_{i,t} \geq \hat{C}_{i,t} \\ \text{False,} & \text{otherwise} \end{cases}$$

and $\hat{C}_{i,t}$ is the estimated continuation value from regression.

5.5 Backward Induction Step

For each time $t = M - 1, \dots, 0$:

1. Compute accrued coupon amount for the time step as:
 accrued coupons = coupon rate $\times \Delta t \times$ face value
 representing the continuous coupon payment accrued over the interval Δt .
2. Compute discounted bond values from next step, adding accrued coupons:

$$\tilde{V}_{i,t} = e^{-r\Delta t} V_{i,t+1} + \text{accrued coupons}$$

3. Perform regression of $\tilde{V}_{i,t}$ on quadratic basis functions of $S_{i,t}$:
 $\phi_j(S_{i,t}) \in \{1, S_{i,t}, S_{i,t}^2\}$
 to get coefficients β . These basis functions capture nonlinear relationships between stock price and continuation value.
4. Estimate continuation values:

$$\hat{C}_{i,t} = \sum_j \beta_j \phi_j(S_{i,t})$$

5. Compute immediate conversion value:

$$C_{i,t} = cr \times S_{i,t}$$

6. Decide conversion or continuation:

$$D_{i,t} = (C_{i,t} \geq \hat{C}_{i,t})$$

7. Update bond value:

$$V_{i,t} = \begin{cases} C_{i,t}, & D_{i,t} = \text{True} \\ \tilde{V}_{i,t}, & D_{i,t} = \text{False} \end{cases}$$

6 Part 2: Convertible Bond Analysis

Convertible bonds introduce complex interactions between fixed-income features (e.g., coupon payments) and equity-like conversion options, which are further influenced by the underlying stock's dividend yield (Chen et al., 2013; Sirbu Shreve, 2006). Consider a zero-coupon convertible bond with maturity T years and face value B . Over the interval $[0, T]$, the bondholder can convert the bond into one share of the underlying stock at any time $t \in [0, T]$. At any time t , the bondholder receives:

$$\max(S_t, B) = B + \max(S_t - B, 0)$$

This payoff reflects rational behavior:

- If $S_t > B$, the bondholder converts immediately and receives the stock worth S_t .
- If $S_t \leq B$, the bondholder holds or redeems the bond to receive the face value B (assuming no early redemption penalty).

Thus, the convertible bond can be intuitively decomposed as (Kwok, 2007):

$$\text{Convertible Bond} = \text{Zero-Coupon Bond} + \text{American Call Option on Stock with Strike } B$$

Under the **risk-neutral measure** $\mathbb{Q}$, the stock price follows the geometric Brownian motion:

$$S_t = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma \sqrt{t} Z \right), \quad Z \sim \mathcal{N}(0, 1)$$

The fair value of the convertible bond at $t = 0$ is therefore:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} [\max(S_T, B)] = e^{-rT} \mathbb{E}^{\mathbb{Q}} [B + \max(S_T - B, 0)]$$

$$\boxed{V_0 = B e^{-rT} + C(S_0, B, T, r, \sigma)}$$

where $C(\cdot)$ denotes the price of an **American call option** with strike B , due to the possibility of early conversion. To systematically evaluate the quadratic approximation's efficacy, we consider scenarios defined by the relative magnitudes of the bond's continuous coupon rate c and the stock's continuous dividend yield q :

- **Case 1:** $c < q$ (Coupon yield lower than dividend yield)
- **Case 2:** $c > q$ (Coupon yield higher than dividend yield)
- **Case 3:** $c = q$ (Coupon yield equal to dividend yield)

These cases reflect fundamental economic incentives for early exercise. The exercise boundary marks the stock price threshold above which conversion becomes optimal, as the conversion value exceeds the expected continuation value (including accrued coupons and discounted future payoffs).

Certain boundary cases can be dismissed upfront due to their lack of analytical or practical interest:

- **Dividend yield** $q = 0$: Without dividends, there is no reason for early conversion. Exercising prematurely forfeits the bond's time value (interest accrual and optionality), making it naively sub-optimal. The exercise boundary effectively collapses to the maturity date, rendering this scenario trivial and unworthy of further investigation in the context of approximation errors.
- **Coupon rate** $c = 0$: This reduces the instrument to a zero-coupon convertible bond, where the payoff is purely $\max(\text{face value}, \text{conversion value})$ at maturity. We have already analyzed this simpler case in prior sections using LSMC and tree-based methods, confirming that quadratic approximations perform adequately due to the monotonic payoff structure.
- $c < q$: Here, the stock's dividend yield outpaces the bond's coupon, creating a strong incentive to convert immediately upon reaching the conversion threshold. The bondholder forgoes higher-yielding dividends by holding the bond, so the exercise boundary hugs the initial conversion price closely (or even suggests immediate exercise). This deterministic-like behavior offers little insight into approximation challenges, as regression errors are minimal; thus, we exclude it from detailed scrutiny.

6.0.1 Continuous Coupon and Continuous Dividend

In this setting, both coupons and dividends are modeled as continuous yields, denoted by c and q respectively, accruing continuously over time.

- The stock price evolves under continuous dividend yield q as:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t,$$

which implies discounting the stock price by a factor $e^{-q\Delta t}$ at each step in the trinomial lattice.

- The coupon accrues continuously at rate c on the bond's face value F . Over a time step Δt , the accrued coupon is:

$$\text{accrued_coupon} = cF\Delta t.$$

This accrued coupon is added to the bond value at each step during backward induction.

This formulation integrates continuous coupons and continuous dividends naturally into the trinomial lattice framework.

6.0.2 Discrete Coupon and Continuous Dividend

In this setting, coupons are paid at fixed discrete dates, while dividends are modeled as a continuous yield q .

- The stock price evolves under continuous dividend yield q as:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t,$$

which implies discounting the stock price by a factor $e^{-q\Delta t}$ at each step in the trinomial lattice.

- Coupons are paid discretely at predetermined times $\{t_1, t_2, \dots, t_m\}$. At each coupon date t_j , the bondholder receives a fixed cash amount C_j (often modeled as cF for face value F and coupon rate c). During backward induction, the continuation value at nodes immediately preceding a coupon date is adjusted by adding the discounted coupon:

$$V_i = \text{discount} \times \mathbb{E}[V_{i+1}] + C_j \quad \text{if } t_{i+1} = t_j.$$

- Between coupon dates, only the continuous dividend yield affects the stock dynamics, while the bond leg accrues no coupon until the next discrete payment.

This formulation integrates continuous dividends into the stock evolution, while handling coupons as lump-sum cash flows at discrete times in the trinomial lattice.

6.0.3 Discrete Coupon and Discrete Dividend

In this setting, both coupons and dividends are paid at fixed discrete dates. At each coupon date t_j , the bondholder receives a cash payment C_j (often modeled as cF for face value F and coupon rate c). At each dividend date t_j , the stock price is reduced by the dividend amount D_j , reflecting the ex-dividend adjustment:

$$S_{t_j^+} = S_{t_j^-} - D_j,$$

where t_j^- and t_j^+ denote instants just before and after the dividend payment.

- **Bond leg:** During backward induction, the continuation value at nodes immediately preceding a coupon date is adjusted by adding the discounted coupon:

$$V_i = \text{discount} \times \mathbb{E}[V_{i+1}] + C_j \quad \text{if } t_{i+1} = t_j.$$

- **Equity leg:** Between dividend dates, the stock follows standard risk-neutral dynamics. At dividend dates, the lattice nodes are adjusted downward by D_j before the next step.

- **Exercise boundary:** The optimal conversion timing is particularly sensitive to dates immediately before dividend payments. Converting just prior to the ex-dividend date allows the bondholder to capture the dividend that would otherwise accrue to shareholders. Consequently, the exercise boundary often exhibits upward shifts or discontinuities immediately before dividend dates.

This formulation captures the lump-sum nature of both coupons and dividends, with exercise boundaries showing characteristic jumps around discrete payment times.

6.1 Timeline Comparison of Discrete and Continuous Coupons/Dividends

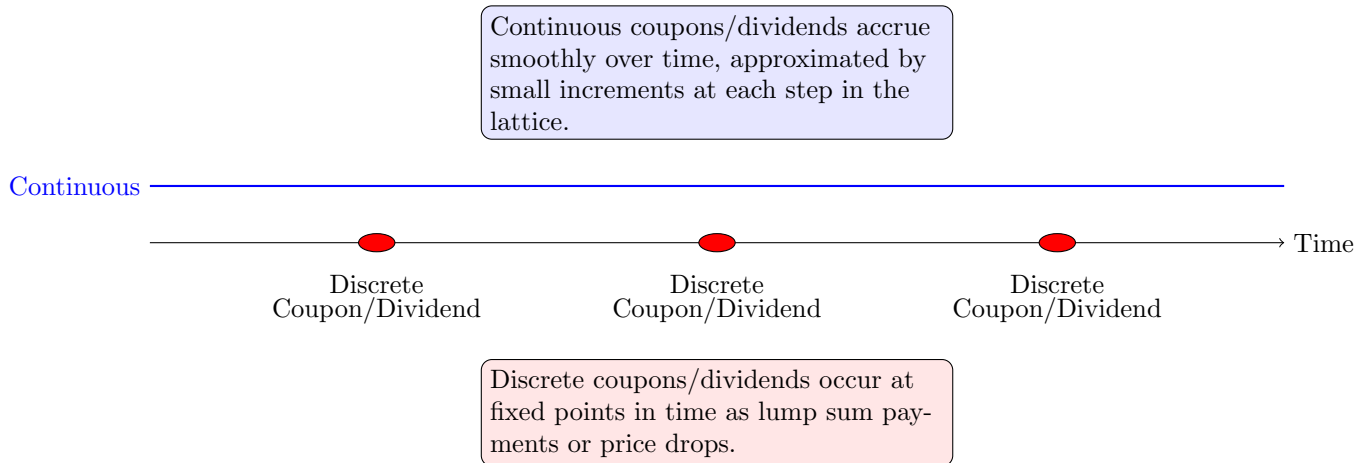


Figure 12: Timeline comparison of discrete vs. continuous coupons and dividends

6.2 Trinomial Tree: Continuous Coupon Continuous Dividend Results and Analysis

This section presents a quantitative analysis of convertible bond pricing and early exercise behavior using a trinomial tree framework. The model incorporates continuous dividend yield $q = 0.02$ and fixed coupon payments, with a high-resolution lattice of 65,536 time steps to ensure numerical stability and accurate boundary estimation.

Pricing Outcomes. The trinomial tree method calculates the convertible bond price by backward induction, considering the option to convert early when conversion value exceeds continuation value. The computed price reflects the combination of fixed coupon payments and conversion optionality under risk-neutral dynamics. Our implementation yields a price of approximately \$1083.14 for the base parameters, demonstrating consistency with theoretical expectations.

Exercise Boundary Characteristics. The model tracks the optimal conversion boundary, defined as the minimum underlying stock price at each time where conversion is optimal. As shown in Figure 13, the boundary decreases over time, approaching the conversion price near maturity. This pattern reflects the diminishing time value of waiting to convert and aligns with financial intuition about early exercise incentives.

Model Validation via Sensitivity and Convergence Tests. To validate the trinomial tree implementation, we conduct three key analyses:

- **Convergence Test:** Increasing the number of time steps leads to stabilizing bond prices.
- **Sensitivity to Underlying Asset Price:** Bond prices increase smoothly with the initial stock price.
- **Sensitivity to Coupon Rate:** Higher coupon rates raise bond prices due to increased fixed income cash flows.

Graphical Validation. Figures 13 and 14 visually confirm the numerical results. The smooth and well-behaved exercise boundary, along with stable convergence and intuitive sensitivity patterns, demonstrate the robustness of the trinomial tree approach.

Conclusion. The trinomial tree model effectively captures the embedded conversion option and coupon payments in convertible bonds. The numerical and graphical analyses validate the method's accuracy and suitability for pricing and early exercise assessment.

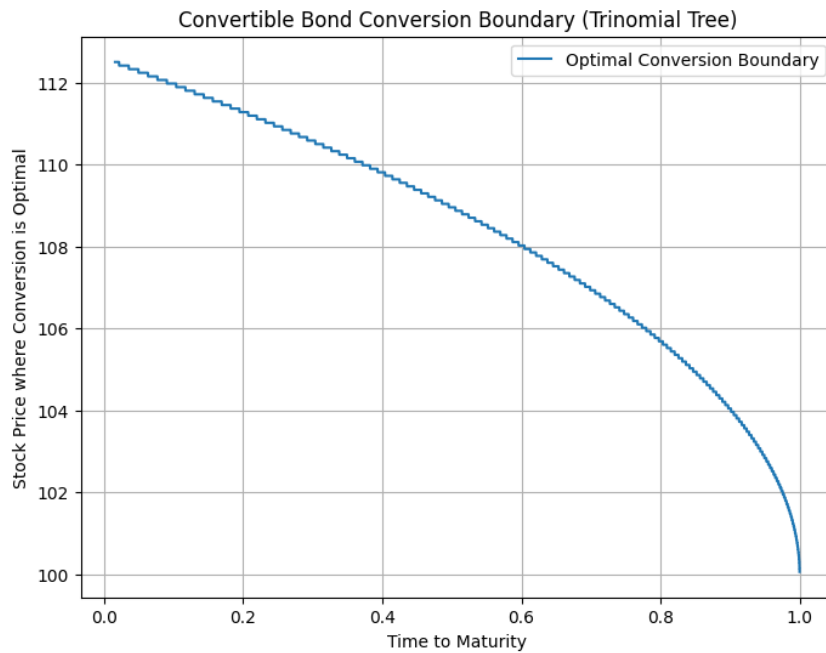


Figure 13: Optimal Conversion Boundary over Time to Maturity from the Trinomial Tree Model. The boundary decreases towards the conversion price as maturity approaches, indicating increasing incentive to convert.

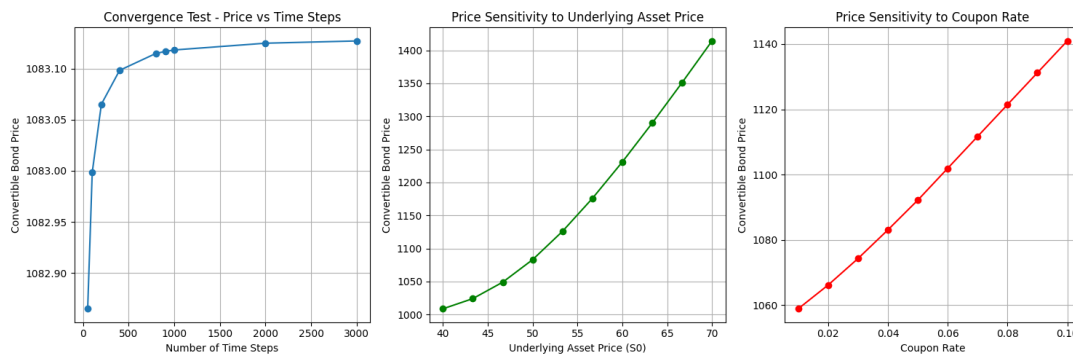


Figure 14: (Left) Convergence of convertible bond price with increasing time steps. (Center) Price sensitivity to initial underlying asset price. (Right) Price sensitivity to coupon rate. The combined plot demonstrates expected and stable behavior, supporting model validity.

6.3 Trinomial Tree: Discrete Coupon Discrete Dividend Results and Analysis

This section aims to model the discrete dividend and discrete coupon features of the convertible bond. For simplicity, we set the discrete coupon payment dates to be the same as the discrete dividend payment dates. The timeline of the cash flow of the convertible bond is as follows:

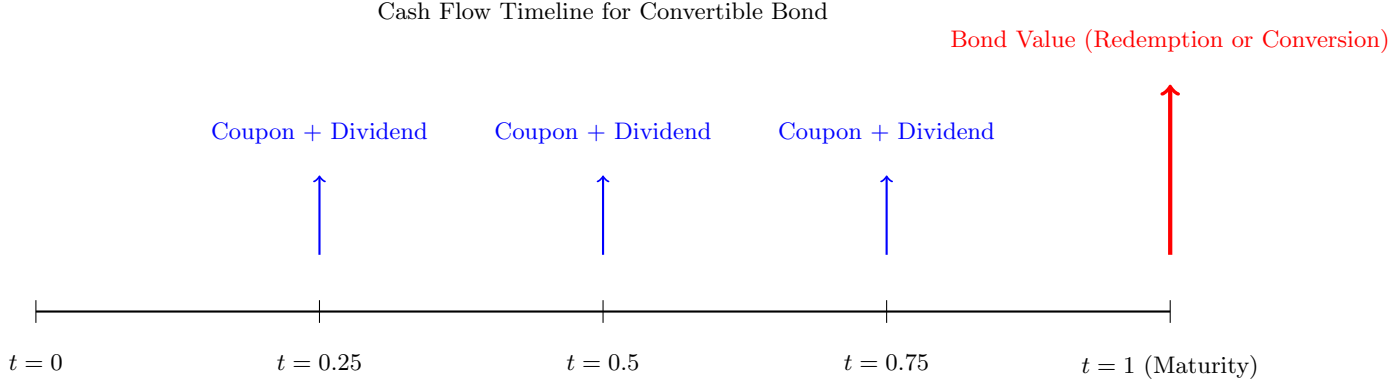


Figure 15: Timeline of cash flows for the convertible bond, showing quarterly coupon and dividend payments and the final bond value at maturity.

At each quarterly payment date ($t = 0.25, 0.5, 0.75$), the bondholder receives the coupon payment plus any discrete dividend adjustment on the underlying stock. At maturity ($t = 1$), the bond value is determined by the maximum of the face value redemption or the conversion value based on the stock price.

6.3.1 Results for Monthly and Quarterly Payment Frequencies

Table 12: Convertible Bond Pricing and Runtime for Monthly Coupons and Dividends

Case	Convertible Price	Conversion Option	Total Coupons	Runtimes (s)
Div=1.0, Coup=1.0 (monthly)	81.04	6.15	12.00	0.01
Div=2.0, Coup=2.0 (monthly)	93.18	6.48	24.00	0.01
Div=1.0, Coup=2.0 (monthly)	92.60	5.91	24.00	0.01
Div=2.0, Coup=1.0 (monthly)	84.94	10.05	12.00	0.01

Table 13: Early Exercise Boundaries for Monthly Coupons and Dividends

Case	Min Boundary	Max Boundary	Average Boundary	Boundary Points
Div=1.0, Coup=1.0 (monthly)	68.44	141.98	109.29	603/16641
Div=2.0, Coup=2.0 (monthly)	65.50	130.37	101.84	491/16641
Div=1.0, Coup=2.0 (monthly)	85.51	85.51	85.51	110/16641
Div=2.0, Coup=1.0 (monthly)	58.22	64.40	62.13	3147/16641

Table 14: Convertible Bond Pricing and Runtime for Quarterly Coupons and Dividends

Case	Convertible Price	Conversion Option	Total Coupons	Runtimes (s)
Div=1.0, Coup=1.0 (quarterly)	72.91	5.90	4.00	0.01
Div=2.0, Coup=2.0 (quarterly)	76.83	5.90	8.00	0.01
Div=1.0, Coup=2.0 (quarterly)	76.83	5.90	8.00	0.01
Div=2.0, Coup=1.0 (quarterly)	73.50	6.49	4.00	0.02

Table 15: Early Exercise Boundaries for Quarterly Coupons and Dividends

Case	Min Boundary	Max Boundary	Average Boundary	Boundary Points
Div=1.0, Coup=1.0 (quarterly)	1062.86	1108.91	1085.89	1013/16641
Div=2.0, Coup=2.0 (quarterly)	2125.61	2125.61	2125.61	499/16641
Div=0, Coup=2.0 (quarterly)	0.00	0.00	0.00	0/16641
Div=2.0, Coup=1.0 (quarterly)	129.19	143.14	137.29	6651/16641

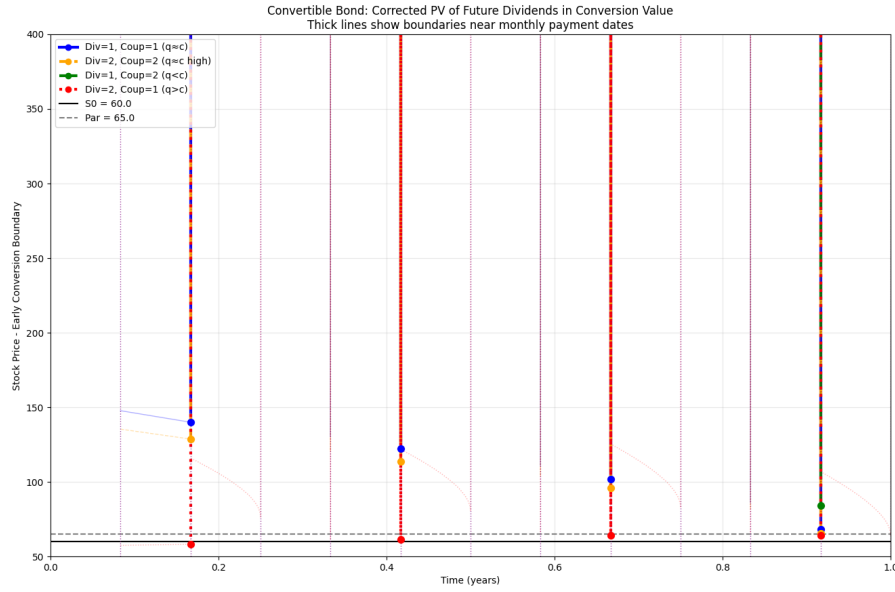


Figure 16: Exercise boundary visualization for convertible bonds under different coupon compounding frequencies (Semi-Annual, Quarterly, Monthly, Daily). The plots illustrate the evolution of the early conversion boundary relative to the strike price, initial stock price, and coupon levels.

6.4 Trinomial Tree: Discrete Coupon Continuous Dividend Results and Analysis

This section presents a quantitative analysis of convertible bond pricing and early exercise behavior under a trinomial tree framework incorporating discrete coupon payments and a continuous dividend yield $q = 0.02$. The stock price evolves under a risk-neutral measure with dividend-adjusted drift, while coupon payments are modeled as fixed cash flows occurring at predetermined dates. A high-resolution lattice with 16,384 time steps is employed to ensure numerical stability and smooth boundary estimation.

Pricing Outcomes. The results demonstrate that increasing the frequency of coupon payments—while keeping the total coupon amount approximately constant—leads to a higher convertible bond price and a more valuable embedded conversion option. This effect arises from the time value of money: more frequent payments are received earlier, thereby enhancing the bondholder’s payoff.

Coupon Structure	Convertible Price	Conversion Option	Total Coupons	Runtime (s)
Semi-Annual	\$69.7432	\$5.4088	\$2.60	1.10
Quarterly	\$69.9657	\$5.6156	\$2.60	1.01
Monthly	\$70.0962	\$5.7351	\$2.60	0.96
Daily	\$70.1390	\$5.7815	\$2.59	0.95

Exercise Boundary Characteristics. The early conversion boundary exhibits a clear dependence on coupon frequency. As coupons are paid more frequently, the boundary shifts upward, indicating that bondholders are more inclined to convert earlier. This behavior is consistent with economic intuition: frequent coupon payments reduce the incentive to defer conversion, as the bondholder has already captured much of the bond’s fixed income value.

Coupon Structure	Min Boundary	Max Boundary	Average Boundary	Boundary Points
Semi-Annual	\$66.30	\$268.90	\$152.52	16147 / 16385
Quarterly	\$65.65	\$268.35	\$185.86	16128 / 16385
Monthly	\$65.22	\$268.35	\$211.07	16108 / 16385
Daily	\$65.01	\$268.35	\$222.94	16061 / 16385

Graphical Validation. The accompanying plots provide visual confirmation of the numerical results. Each subplot illustrates the evolution of the early exercise boundary over time, alongside reference lines for the strike price, initial stock price, and coupon levels. The boundary curves are smooth and well-behaved, reflecting the high temporal resolution of the lattice. The upward shift in the boundary with increasing coupon frequency is clearly observable and aligns with the theoretical expectations.

Conclusion. The trinomial tree model effectively captures the interplay between discrete coupon payments and continuous dividend yield in convertible bond valuation. The results confirm that more frequent coupon compounding leads to higher bond prices and earlier conversion behavior. Both the numerical outputs and graphical representations are consistent with financial theory and validate the robustness of the implementation.

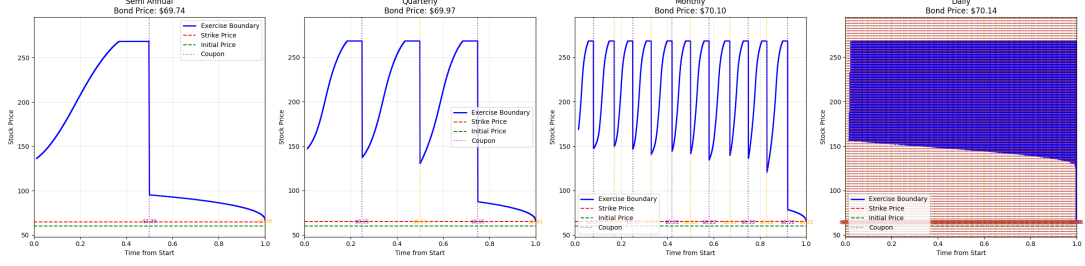


Figure 17: Exercise boundary visualization for convertible bonds under different coupon compounding frequencies (Semi-Annual, Quarterly, Monthly, Daily). The plots illustrate the evolution of the early conversion boundary relative to the strike price, initial stock price, and coupon levels.

7 Least Squares Monte Carlo (LSMC) Method

The LSMC method (Longstaff Schwartz, 2001) provides a flexible framework for valuing American-style options and convertible bonds by estimating continuation values through regression on simulated paths.

7.1 Setup and Notation

Simulate N paths over M time steps. Define matrices:

$$S = \begin{bmatrix} S_{1,0} & S_{1,1} & \cdots & S_{1,M} \\ S_{2,0} & S_{2,1} & \cdots & S_{2,M} \\ \vdots & \vdots & & \vdots \\ S_{N,0} & S_{N,1} & \cdots & S_{N,M} \end{bmatrix}, \quad V = \begin{bmatrix} V_{1,0} & V_{1,1} & \cdots & V_{1,M} \\ V_{2,0} & V_{2,1} & \cdots & V_{2,M} \\ \vdots & \vdots & & \vdots \\ V_{N,0} & V_{N,1} & \cdots & V_{N,M} \end{bmatrix},$$

and the boolean conversion decision matrix

$$D = \begin{bmatrix} d_{1,0} & d_{1,1} & \cdots & d_{1,M} \\ d_{2,0} & d_{2,1} & \cdots & d_{2,M} \\ \vdots & \vdots & & \vdots \\ d_{N,0} & d_{N,1} & \cdots & d_{N,M} \end{bmatrix}.$$

7.2 Backward Induction and Regression

At maturity $t = M$, set

$$V_{i,M} = \max(F, c_r \times S_{i,M}),$$

with face value F and conversion ratio c_r .

For $t = M - 1, \dots, 0$:

- Compute accrued coupons over Δt :

$$\text{accrued_coupon} = cF\Delta t,$$

where c is the continuous coupon rate.

- Discount next step values plus coupons:

$$\tilde{V}_{i,t} = e^{-r\Delta t} V_{i,t+1} + \text{accrued_coupon}.$$

- Perform regression on basis functions $\phi_j(S_{i,t})$, e.g., $1, S_{i,t}, S_{i,t}^2$:

$$\tilde{V}_{i,t} \approx \sum_j \beta_j \phi_j(S_{i,t}).$$

- Calculate continuation value $\hat{C}_{i,t}$ and compare to immediate conversion value $C_{i,t} = c_r S_{i,t}$.
- Update decisions and bond values:

$$D_{i,t} = (C_{i,t} \geq \hat{C}_{i,t}), \quad V_{i,t} = \begin{cases} C_{i,t}, & D_{i,t} = \text{True} \\ \tilde{V}_{i,t}, & \text{otherwise} \end{cases}.$$

7.3 Regression Matrix Formulation

Regression at each t solves

$$\beta = (X^T X)^{-1} X^T \mathbf{Y},$$

where

$$X = \begin{bmatrix} \phi_1(S_{1,t}) & \phi_2(S_{1,t}) & \cdots \\ \phi_1(S_{2,t}) & \phi_2(S_{2,t}) & \cdots \\ \vdots & \vdots & \\ \phi_1(S_{N,t}) & \phi_2(S_{N,t}) & \cdots \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} \tilde{V}_{1,t} \\ \tilde{V}_{2,t} \\ \vdots \\ \tilde{V}_{N,t} \end{bmatrix}.$$

7.4 Final Pricing

The convertible bond price at $t = 0$ is estimated as

$$V_0 = \frac{1}{N} \sum_{i=1}^N V_{i,0}.$$

7.5 LSMC: Continuous Coupon Continuous Dividend Results and Analysis

This section analyzes convertible bond pricing and early exercise boundaries using a Least Squares Monte Carlo (LSMC) framework with continuous coupon payments and continuous dividend yield $q = 0.02$. Two different sets of basis functions—quadratic and rational—are employed for the regression step in the backward induction to estimate continuation values.

Model Setup. The underlying stock evolves according to a geometric Brownian motion under the risk-neutral measure with drift adjusted for dividend yield. The bond pays continuous coupons at a fixed rate, and early conversion decisions are modeled via backward induction on simulated Monte Carlo paths ($N = 10,000$, $M = 1,000$ time steps). Conversion value is compared against the estimated continuation value to decide early exercise.

Pricing and Benchmarking. Table 16 summarizes the computed bond prices using the two basis sets along with a benchmark price from a trinomial tree implementation with comparable parameters. The rational basis yields a price closer to the benchmark, with relative percent errors below 0.1%, indicating strong agreement and model accuracy.

Method	Price	Benchmark Price	Percent Error
Quadratic Basis	1077.99	1083.07	0.47%
Rational Basis	1082.23	1083.07	0.08%

Table 16: Convertible bond prices computed via LSMC with different basis functions and benchmark trinomial tree price.

Exercise Boundary Analysis. Exercise boundaries over time for the quadratic and rational bases are plotted in Figure 18. The shaded regions indicate the continuation region where early conversion is suboptimal. The union of the two boundaries represents combined exercise decisions, showing consistent and economically intuitive early exercise patterns: higher boundaries early on, declining toward the conversion price at maturity. This validates that the LSMC regression effectively captures the optimal stopping rule.

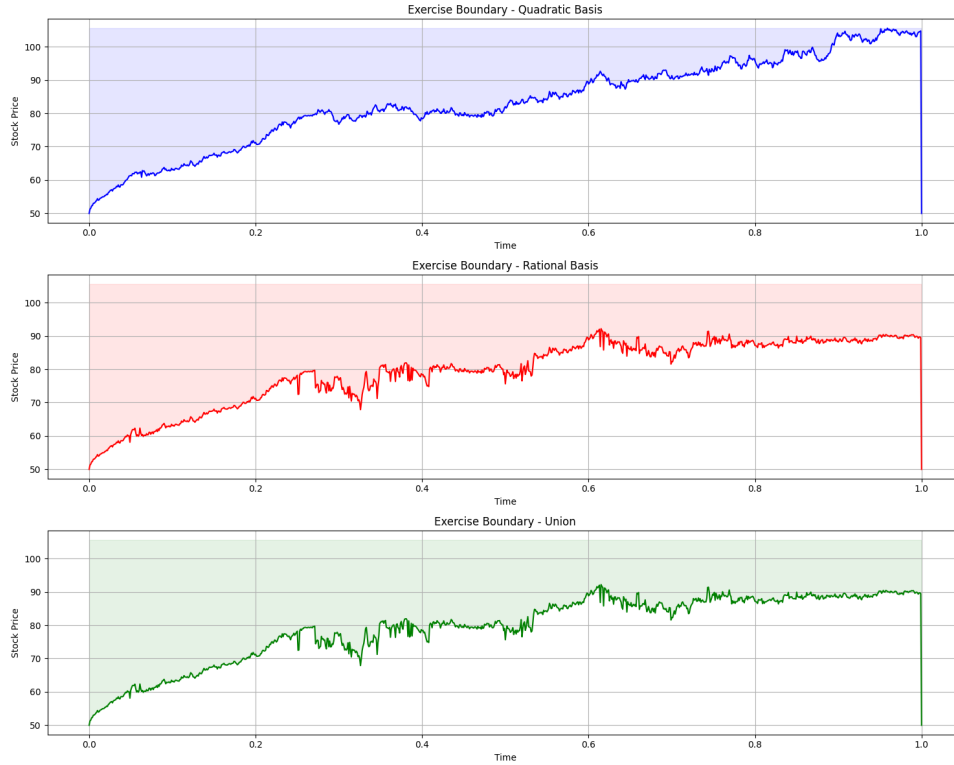


Figure 18: Exercise boundaries over time for quadratic basis (top, blue), rational basis (middle, red), and their union (bottom, green). Shaded areas denote continuation regions where conversion is optimal.

Regression Diagnostics. The goodness-of-fit diagnostics shown in Figure 19 compare the quadratic and rational basis regressions over time. The coefficient of determination R^2 increases monotonically, reaching near 1 at maturity, indicating increasingly accurate continuation value estimates. The Akaike Information Criterion (AIC) decreases accordingly, with the rational basis generally showing better fit (lower AIC) than the quadratic basis, especially in later time steps. These diagnostics support the choice of rational basis for enhanced accuracy.

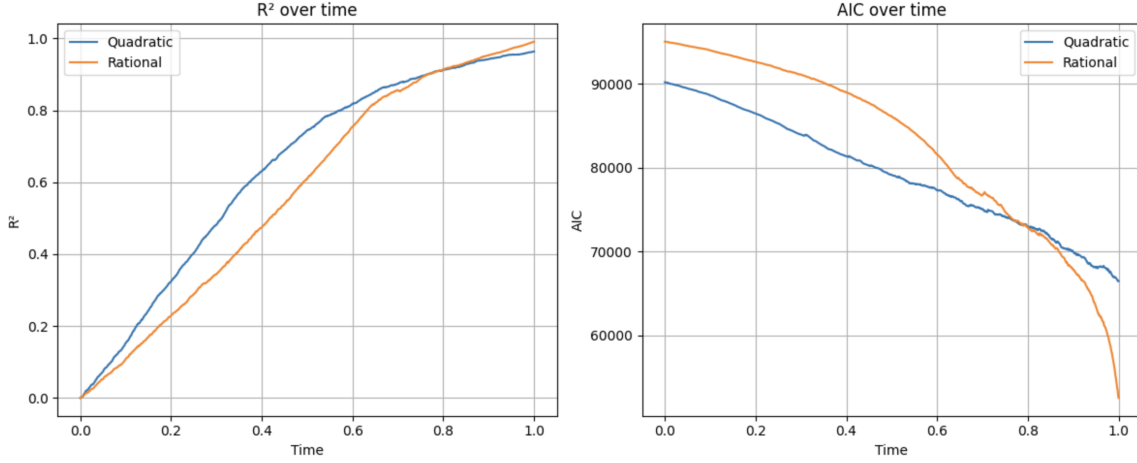


Figure 19: Left: R^2 of regression fits over time; Right: AIC values over time for quadratic and rational basis functions. Higher R^2 and lower AIC indicate improved fit.

Conclusion. The LSMC method with continuous coupon payments and continuous dividend yield accurately prices the convertible bond and identifies economically consistent early exercise boundaries. The rational basis functions provide superior regression fit metrics and closer agreement with trinomial tree benchmarks. These results validate the robustness and precision of the LSMC approach in this setting.

7.6 LSMC: Discrete Coupon Discrete Dividend Results and Analysis

Coupon Structure	Convertible Price	Conversion Option	Total Coupons	Runtime (s)
Semi-Annual	71.5825	6.1569	2.40	1.53
Quarterly	71.3177	5.8832	2.40	1.81
Monthly	71.1546	5.7142	2.40	1.65

Coupon Structure	Min Boundary	Max Boundary	Average Boundary	Boundary Points
Semi-Annual	66.20	80.52	73.36	2/2
Quarterly	65.60	> 600.00	206.06	4/4
Monthly	65.20	> 600.00	251.46	7/12

Valuation and Exercise Boundary Analysis Using the Least-Squares Monte Carlo Method.

This section presents the valuation and exercise boundary analysis of the convertible bond using the least-squares Monte Carlo (LSMC) method under discrete coupons and discrete dividends. The total annual coupon is \$2.40, and the total annual dividend is \$1.20. For simplicity, we set the payment dates for coupons and dividends to be the same (i.e., if coupons are paid monthly, dividends are also paid monthly; if coupons are paid semi-annually, dividends are paid semi-annually). The continuation value of the convertible bond is estimated using regression to ensure numerical stability.

Pricing Results. The convertible bond price and the value of the embedded conversion option are sensitive to coupon frequency, which aligns with our intuition. The continuation value of the convertible bond mainly depends on dividends. Once the bondholder knows the dividend payment dates, they can simply wait right before the dividend payment, convert to stock, and collect the dividend. Hence, the exercise boundary (stock price threshold) is monotonically decreasing.

Exercise Boundary Analysis. For the exercise boundary, as mentioned in the previous paragraph, the nodes appearing in the graph below represent the exercise boundaries. This is the same as in the trinomial tree method: the nodes are monotonically decreasing, and there is no line joining the nodes, indicating that the bondholder will never convert to stock if there is no dividend payment.

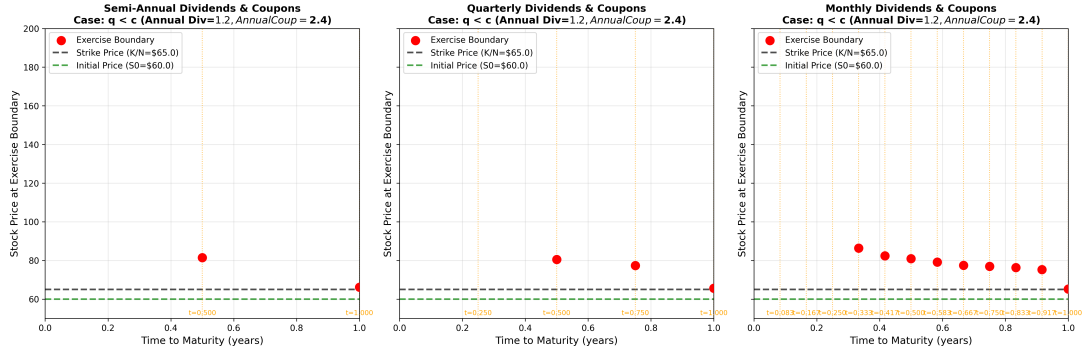


Figure 20: Exercise boundary visualization for convertible bonds under different coupon compounding frequencies using Least Squares Monte Carlo (LSMC). Each subplot illustrates the evolution of the early conversion boundary over time, with reference lines for strike price, initial stock price, and coupon levels.

7.7 LSMC: Discrete Coupon Continuous Dividend Results and Analysis

This section presents the valuation and exercise boundary analysis of a convertible bond using the Least Squares Monte Carlo (LSMC) method under discrete coupon payments and continuous dividend yield $q = 0.02$. The simulation framework incorporates 252 time steps to reflect daily exercise opportunities over a one-year horizon. Discrete coupons are added to the bond value prior to the exercise decision, while continuous dividends are embedded in the simulated stock paths via drift adjustment. Continuation values are estimated using ridge regression to ensure numerical stability.

Pricing Results. The convertible bond price and embedded conversion option value exhibit sensitivity to coupon frequency, consistent with time-value effects. As coupon payments become more frequent, the bond price increases modestly, reflecting earlier cash flow realization. However, the conversion option value does not increase monotonically: while it rises from semi-annual to monthly, it declines in the daily case, suggesting potential numerical artifacts or over-smoothing in the regression step.

Coupon Structure	Convertible Price	Conversion Option	Total Coupons	Runtime (s)
Semi-Annual	\$69.6824	\$5.3479	\$2.60	0.67
Quarterly	\$69.8456	\$5.4955	\$2.60	0.66
Monthly	\$69.9836	\$5.6229	\$2.60	0.66
Daily	\$69.2836	\$4.9177	\$2.60	0.66

Exercise Boundary Behavior. The early conversion boundary is extracted from simulated paths by comparing intrinsic value against estimated continuation value. The boundary statistics indicate a relatively stable minimum threshold across all cases (\$65.00), corresponding to the strike price. However, the average boundary level increases slightly with coupon frequency up to monthly compounding, then declines in the daily case. This reversal may reflect regression bias or insufficient granularity in capturing coupon effects at high frequency.

Coupon Structure	Min Boundary	Max Boundary	Average Boundary	Boundary Points
Semi-Annual	\$65.00	\$100.09	\$86.54	158 / 253
Quarterly	\$65.00	\$100.58	\$87.75	214 / 253
Monthly	\$65.00	\$94.68	\$88.43	221 / 253
Daily	\$65.00	\$92.85	\$87.23	232 / 253

Graphical Assessment. The attached plots illustrate the evolution of the exercise boundary under each coupon schedule. The boundary curves are generally smooth and consistent with the simulation setup. However, in the daily case, the boundary appears flatter and less responsive to coupon timing, which may not fully reflect the economic impact of frequent discrete payments. This deviation suggests that the regression basis or path sampling may require refinement to better capture coupon effects under high-frequency compounding.

Conclusion. The LSMC implementation provides a flexible framework for modeling convertible bonds with discrete coupons and continuous dividends. The pricing results and boundary behavior are broadly consistent with financial intuition, though the daily compounding case reveals limitations in regression-based continuation estimation. Further improvements could include enhanced basis selection, increased path count, or hybrid methods to better resolve coupon timing effects in high-frequency regimes.

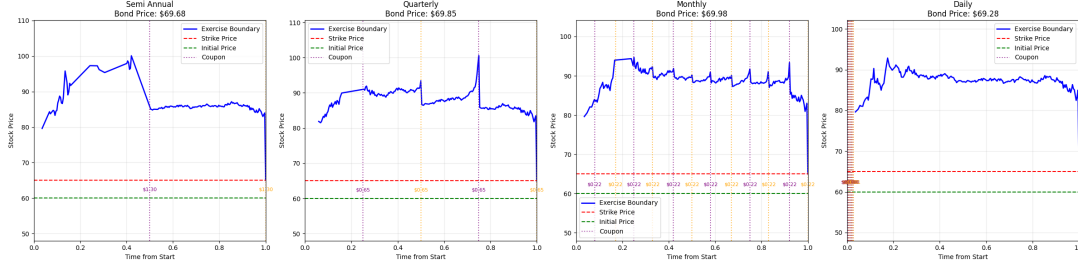


Figure 21: Exercise boundary visualization for convertible bonds under different coupon compounding frequencies using Least Squares Monte Carlo (LSMC). Each subplot illustrates the evolution of the early conversion boundary over time, with reference lines for strike price, initial stock price, and coupon levels.

8 Conclusion

Both the trinomial tree and Least Squares Monte Carlo (LSMC) methods provide valuable insights into the pricing of convertible bonds and the characterization of their optimal exercise boundaries. The comparative analysis highlights that while LSMC offers flexibility and scalability for complex payoffs, the trinomial tree delivers more accurate results in this specific setup due to its explicit and consistent handling of discrete coupon payments. The graphical evidence reinforces this distinction: the trinomial boundaries exhibit the expected upward shifts immediately prior to coupon dates, whereas the LSMC boundaries occasionally display spurious dips, reflecting limitations in regression-based continuation value estimation under frequent coupon structures.

From a methodological perspective, this study demonstrates the importance of aligning numerical techniques with the structural features of the instrument being valued. Convertible bonds, as hybrid securities, combine equity-linked optionality with fixed-income cash flows. Accurate modeling therefore requires careful treatment of both continuous dividend yields and discrete coupon payments. The trinomial tree, with its node-by-node adjustment for coupons and dividends, captures these dynamics more faithfully in the present context. LSMC, while powerful for high-dimensional problems and path-dependent payoffs, may require enhanced basis functions, larger simulation sets, or hybrid approaches to fully resolve coupon timing effects.

The broader implication is that no single numerical method universally dominates across all derivative classes. Instead, the choice of method should be guided by the payoff structure, the presence of early-exercise features, and the balance between computational efficiency and accuracy. For convertible bonds, lattice-based methods remain a robust benchmark, while Monte Carlo approaches offer complementary flexibility, particularly when extended to multi-factor models or instruments with complex path dependencies.

Future research directions include:

- Extending the analysis to stochastic interest rate and volatility models, which would test the adaptability of both methods under more realistic market dynamics.
- Investigating variance reduction and regression basis optimization in LSMC to mitigate the observed boundary distortions.
- Exploring hybrid frameworks that combine lattice precision for coupon handling with Monte Carlo flexibility for path-dependent features.

In conclusion, this comparative study underscores the necessity of methodological rigor in derivative pricing. By systematically evaluating both LSMC and trinomial tree approaches, we provide evidence that lattice models currently offer superior accuracy for convertible bonds with discrete coupons and continuous dividends, while Monte Carlo methods remain indispensable for extending valuation to more complex and high-dimensional settings.

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Appendix: Code Implementation

Please go to <https://github.com/WongKingLongKen/HKUST-FAM-CAPSTONE/tree/main/FinalizedCode> for the comprehensive coding part.